

No additional strata, no loop! - A re-analysis of Malayalam compound morphophonology

Abstract: In this article, we present an analysis of the morphophonological properties of Malayalam compounds, with a particular focus on stress. Our analysis maintains the original claim of strict level ordering made in Lexical Phonology and Morphology (LPM) (Kiparsky 1982), which ensures a restrictive interleaving of morphology and phonology. The morphophonological behaviour of Malayalam compounds is first presented and analysed by Mohanan (1986) in LPM. He claims that different compound types are associated with distinct strata due to systematic phonological differences between compounds. However, compound embedding indicates that these strata must interact bidirectionally, which necessitates the introduction of a stratal loop. Introducing the loop means giving up strict ordering of strata, which results in a less restrictive architecture of grammar. Contrary to Mohanan (1986), we show that the loop is an unnecessary tool once the assumption is made that all compounding takes place in a single stratum. We explain the phonological differences in compounds by a difference in prosodic structure, thereby replacing the stratal distinction. Further, our analysis extends naturally to two cases of opaque phonology. Mohanan (1996) claims that both cases provide irrefutable evidence for the stratal distinction of compounds. However, we show that both opaque interactions follow from the proposed prosodic difference in compound types and strictly ordered strata.

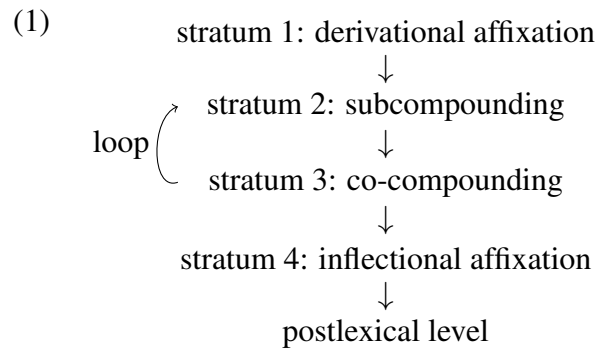
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1 Introduction

Traditional Lexical Phonology and Morphology (LPM) (Kiparsky 1982, Mohanan 1986) and its modern counterpart Stratal Optimality Theory (OT) (Kiparsky 2000, Bermúdez-Otero 2018) assume an architecture of grammar where morphological and phonological rules¹ are distributed across strata. These strata are assumed to be strictly ordered (Kiparsky 1982, Mohanan 1986), which implies that if stratum 2 is ordered after stratum 1, then morphological and phonological rules of stratum 2 cannot precede those of stratum 1. Mohanan (1986) visualises the assumption of strict ordering with a metaphor

that treats the lexicon as a word-making factory. In this factory, a word is built via a conveyor belt which leads through multiple rooms, the strata. In each room, distinct morphological (affixation, compounding etc.) and phonological rules apply to the word. Crucially, the word can travel on the conveyor belt from room 1 to room 2, but cannot traverse backwards from room 2 to room 1; the conveyor belt goes only in one direction. However, Mohanan shows that to account for the morphophonology of compounds in Malayalam, and in particular stress in compounds, the one-way directionality of the “conveyor belt” cannot be upheld. He asserts that Malayalam requires a looping mechanism to allow the derivation to revisit an earlier stratum. In this paper, we show that strict ordering in a stratal architecture can be maintained by merging Mohanan’s (1986) two compound strata into a single stratum, thereby getting rid of the loop. This counters previous claims on the invalidity of stratal ordering (Orgun 1996, Inkelas 2014) and provides a strong argument to maintain a restrictive approach to morphological and phonological interleaving. Our analysis of compound stress in Malayalam follows Sproat (1986) and Inkelas (1989) who replace Mohanan’s stratal distinction with a prosodic distinction which obviates the need for a loop. Further, we expand on Sproat and Inkelas by covering the entire empirical domain of compound stress and its opaque interactions with the phonological system of Malayalam (Mohanan 1996). We maintain that the attested phonological opacity neither requires the separation of compound types into distinct strata nor the loop. Instead, opacity is a reflex of prosodic structure and strict stratal ordering.

The starting point of the present paper is the analysis presented in Mohanan (1982, 1986) of the morphology and phonology of compounds in Malayalam. Alongside Pesetsky (1979) and Kiparsky (1982), Mohanan’s (1986) work is attributed to the creation and development of LPM. Mohanan provides an analysis of Malayalam’s morphophonology which makes use of four morphophonologically distinct lexical strata, as well as a postlexical stratum. In addition, he assumes a loop from stratum 3 back to stratum 2. His stratal architecture is presented in (1).



The motivation to distinguish four lexical strata, and in particular to distinguish subcompounding (*tatpuruṣas*) from co-compounding (*dvandva*), comes from a difference in their phonology. Example (2) shows the compounding of two nouns *box* and *grainbin*. In (2a), the two nouns form a subcompound, which entails a structure such that *grainbin* is the head of the compound and *box* is its modifier. In (2b), the two nouns form a coordinative compound. Different to subcompounds, a co-compound has a structure where both nouns are treated as heads, which leads to a symmetric, coordinative interpretation (see Wälchli 2005, Ralli 2019 for an overview on coordinative compounds). The two types of compounds differ not only in their morphological structure but also in their surface phonology. Firstly, the two compound types surface with different stress domains. Subcompounds surface as one stress domain, indicated by the bracketing, while co-compounds surface as two stress domains². Secondly, subcompounds undergo lengthening processes that co-compounds do not. In (2a), a derived geminate consonant, boxed for convenience, surfaces at the subcompound boundary. In (2b) no geminate surfaces in the co-compound.

- (2) a. [(pètti^{pp}attá:jám)ɳalə]
 petti- patta:jam -kaɭ
 box- grainbin -PL
 ‘grainbins used as boxes’ [subcompound] (p.90)³

- b. [(pèttí)(p_{nn}attà:jám)ŋalə]
petti- patta:jam -ka
box- grainbin -PL
‘boxes and grainbins’ [co-compound] (p.90)

Mohanan takes these phonological differences to mean that sub- and co-compounds undergo distinct phonological rules, which in LPM means that the two compound types are built at distinct morphological strata. He proposes that nouns subcompound at stratum 2, where they undergo the rule of stress assignment as a compounded unit. Here, a subcompound also undergoes other rules, such as stem initial consonant gemination. Nouns that co-compound only do so at the next stratum, at stratum 3, where rules such as stem initial consonant gemination are turned off and stress assignment has already applied to the individual nouns at stratum 2, prior to stratum 3 co-compounding. This analysis is sketched below in (3) with the examples of (2).

(3) *Stratal derivation of Malayalam compound morphophonology in Mohanan (1986):*

	subcompound	co-compound
stratum 2 morphology		
subcompounding	[[petti] [p _{nn} attà:jam]]	N/A [petti] [p _{nn} attà:jam]
stratum 2 phonology		
initial consonant gemination	[[petti] [pp _{nn} attà:jam]]	N/A
stress assignment	[[pètti] [pp _{nn} attà:jám]]	[pèttí] [p _{nn} attà:jám]
stratum 3 morphology		
co-compounding	N/A [pèttipp _{nn} attà:jám]	[[pèttí] [p _{nn} attà:jám]]
final result:	[pèttipp _{nn} attà:jámŋalə]	[pèttíp _{nn} attà:jámŋalə]

However, contrary to what the phonology indicates, the morphology suggests that both compound types are built at a single morphological stratum. Evidence for this comes from compound embedding. The examples in (4) show a subcompound em-

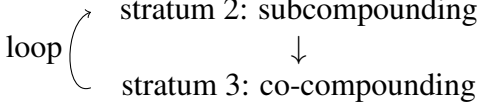
bedded in a co-compound in (4a) and a co-compound embedded in a subcompound in (4b). In the remainder of this paper, we refer to embedded compounds as complex compounds.

- (4) a. [[kà:mukí: [b^hà:ɽjá:sahó:daɽí]] ma:ɽə]
 ka:muki- b^ha:ɽja- saho:daɽi -ma:ɽ
 lady.love- wife- sister -PL
 ‘lady love and wife’s sister’ [co-compound[subcompound]] (p.123)

- b. [[jà:ɽímàtá] wiɽwè:ɽám]
 ja:ɽi- maɽam- wiɽwe:ɽam
 caste- religion- hatred
 ‘hatred of religion and caste’ [[co-compound]subcompound] (p.123)

The possibility for bidirectional embedding as in (4) implies that the two types of compounds interact morphologically. Therefore, word formation of the two compounds should apply at the same morphological stratum.

Mohanan himself recognises that the phonological and morphological evidence is in conflict with respect to the stratal association of compounds. In order to accurately reflect that the phonologies between sub- and co-compounds differ while retaining the possibility for bidirectional embedding, he introduces a stratal loop, repeated as an excerpt of (1) in (5) below.

- (5)
- 
- stratum 2: subcompounding
 ↓
 stratum 3: co-compounding

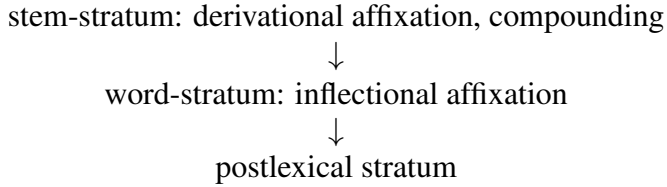
The stratal architecture in (5) derives that subcompounds are phonologically distinct from co-compounds by associating them to different strata. Simultaneously, (5) allows both types of embedding by virtue of the loop. However, the loop is a direct contradiction to the strict ordering of strata, which is argued to be an integral part of

LPM (Kiparsky 1982). Essentially, the loop allows morphological and phonological rules of a later stratum to apply before those of an earlier stratum, deemed impossible under strict ordering. Yet, Malayalam seems to require the loop for the particularly troublesome case in (4b) where a co-compound is embedded in a subcompound. In this construction, the loop allows the grammar to first build a co-compound at stratum 3 and then loop back to stratum 2 where a subcompound can embed the co-compound.

Immediately after its introduction, the loop received criticism for weakening the theory of LPM (Sproat 1985) and a handful of re-analyses attempt to tackle the complex morphophonology of Malayalam compounds, see Sproat (1986) and Inkelas (1989). These re-analyses can explain the part of Malayalam morphophonology in (2) by doing away with the loop and, instead, introducing a prosodic difference to the phonology of compounds. However, Mohanan's (1996) reply to these re-analyses makes it clear that they fail to derive intransparent stress domains in complex compounds, as well as two cases of phonological opacity in co-compounds which do not occur in subcompounds. Thus, Mohanan (1996) argues that the loop remains essential for Malayalam compound morphophonology and, therefore, must be an integral part of the theory of LPM. In the rest of this article, we, in turn, reply to Mohanan (1996) and expand on Sproat (1985) and Inkelas (1989) by arguing that there is no need for a loop once the compounds are no longer distinguished stratally. We argue that the distinct phonological behaviour, as presented in (2), can be explained by a difference in prosodic structure, which follows from the different morphological structures of the compounds. This assumption allows for a collapse of Mohanan's (1986) stratum 2 and 3 into a single stratum where bidirectional compound embedding is no longer an issue. Further, we tackle the problematic phonology of complex compounds, as well as phonological opacity in co-compounds. We show that despite Mohanan's (1996) claim that these cases provide irrefutable evidence for the loop, both cases can be derived without it, as long as prosodic structure is taken into account and strict ordering of strata is maintained. We couch the analysis into a stratal architecture following Bermúdez-Otero (2018) *pace* Kiparsky (1982) that

consists of a stem-, word- and a postlexical stratum. The proposed strata and their strict order is shown in (6).

(6) *New stratal architecture of Malayalam:*



The rest of the article is structured as follows. Section 2 introduced the stress properties of Malayalam. Section 2.1 deals with deriving the stress domains in simplex compounds, and section 2.2 expands the analysis to complex compounds. In both sections, the prosodic difference which we argue leads to all phonological differences observed in both compound types, is derived from the compounds' distinct morphological structure. Section 3 introduces two instantiations of phonological opacity that arise only in co-compounds. Again, we show that this phonological effect is a reflex of the different prosodic structures of the compound types, as well as an instantiation of a cycle realised by the minimized and strictly ordered stratal architecture in (6).

2 Stress in Malayalam

Mohanan (1986) shows that Malayalam has a stress system with unconventional correlates. Primary stress correlates with a low pitch, indicated here by an accent grave $\grave{\text{v}}$. Primary stress falls in the first syllable, see (7a), except if the first syllable is short and the second syllable is long. In that case, shown in (7b), primary stress falls on the second syllable⁴. Secondary stress correlates with an H pitch and is realised on any long vowel that follows the primarily stressed syllable, see (7c). This secondary pitch is indicated by the accent aigu $\acute{\text{v}}$. Finally, a H pitch is realised on the final syllable of each word in (7). We treat this final H pitch as a H boundary tone. It is indicated with the accent aigu $\acute{\text{v}}$, analogous to secondary stress.

- (7) a. [pàɾatí]
 ‘searched’
- b. [paɾà:tí]
 ‘complaint’ (p.112)
- c. [pà:ɾá:jaɾám]
 ‘reading’ (p.113)

2.1 Stress domains in simplex compounds

As mentioned in section 1, sub- and co-compounds supply different domains for stress, see (7) repeated in (8) below. A subcompound in (8a) is treated as a single domain for stress whereas, each head in a co-compound in (8b) is treated as a single domain for stress. The domains are revealed by the single primary stress in the subcompound versus a single primary stress on the respective heads of the co-compound. In both cases, the inflectional affixes are not part of the stress domain; they do not surface with the H boundary tone. Instead, the H boundary tone lands on the final syllable of the compound. It follows that the domain for stress in subcompounds is the entire compound, whereas, in co-compounds, the domains for stress are the individual nouns, excluding inflectional affixes in both compound structures.

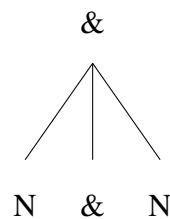
(8) *Stress domain differences in subcompounds versus co-compounds:*

- a. [(pèt̪t̪ippattá:jám)ɳalə]
 pet̪ti- pattá:jam -kaɭ
 box- grainbin -PL
 ‘grainbins used as boxes’ [subcompound](p.90)

- b. [(pèt[í](pattà:ján)ɲalə]
pet[ti]- patta:jam -ka[
box- grainbin -PL
‘boxes and grainbins’ [co-compound](p.90)

We argue that what differs between sub- and co-compounds is not their association to distinct morphological strata as Mohanan (1986) suggests. Instead, what differs is the morphological structure of sub- and co-compounds, which maps onto distinct prosodic structures that impose distinct stress domains. In detail, we propose that co-compounds have the morphological structure in (9a), a balanced nominal compound headed by an abstract & (see Di Sciullo 2005, 2011 for arguments to assume a functional head AND (and others) in compounds and Nóbrega 2020 to assume an abstract & head in co-compounds). This structure leads to the prosodic parse in (10a), where both nouns are parsed as their own separate prosodic words. Subcompounds have the morphological structure in (9b), a balanced nominal compound structure. This structure parses into a single prosodic word encompassing both nominal elements in the compound structure in (10b). To exclude inflectional affixes from the stress domain, we simply assume that they are word-stratum affixes. They are therefore introduced in the morphology only after compounds have been built and stress has been assigned at the stem-stratum.

- (9) a. *Morphological structure of a co-compound:* b. *Morphological structure of a subcompound:*



- (10) a. *Parsing of a co-compound:* b. *Parsing of a subcompound:*
(ω N) (ω N) (ω N N)

To arrive at the parses in (10) from the morphological structures in (9), we adopt MATCH constraints (Selkirk 2011) that match morphosyntactic structure to prosodic structure. Further, we base the computation in Harmonic Grammar (Legendre, Miyata & Smolensky 1990), where constraints are weighted and not ranked. This choice is motivated by a ganging-up effect in the prosodic parsing of complex compounds in section 2.2. The relevant constraints that are needed to arrive at the parses in (10) are presented in (11). The first constraint in (11a) constitutes a typical MATCH constraint. We diverge from Selkirk (2011) in directly referencing the nominal category, as opposed to referencing to an abstract word X that should coincide with the edges of a prosodic word ω . The constraint $\text{MATCH}(N, \omega)$ in (11a) has a comparatively low weight due to the fact that it is violated in the parsing of subcompounds. Constraint (11b) is penalised by parsing recursive prosodic structure. Constraint (11c) is the phonological counterpart to constraint (11a) and is penalised by a prosodic word ω whose edges do not match with the edges of an N . This constraint is relevant for the prosodic parse of co-compounds. Finally, constraint (11d) is penalised if a maximal N is not aligned with a prosodic word ω . We define a maximal N as an N which is not immediately dominated by another N . This constraint is relevant for the prosodic parse of subcompounds and is never violated in Malayalam.

- (11) a. $\boxed{\text{MATCH}(N, \omega)}$: Assign a violation for every constituent type N whose left and right edges in the input morphological representation do not correspond to the left and right edges of a constituent of type ω in the output phonological representation. weight = 4
- b. $\boxed{\text{NONREC}}$: Assign a violation for every prosodic constituent X that dominates a prosodic constituent Y , such that $X=Y$ (compare to Selkirk 1995). weight = 9
- c. $\boxed{\text{MATCH}(\omega, N)}$: Assign a violation for every constituent type ω whose left and right edges in the output phonological representation do not correspond

to the left and right edges of a constituent of type N in the input syntactic representation. weight = 10

- d. $\text{MATCH}(\text{N}^{\text{max}}, \omega)$: Assign a violation for every constituent of type N^{max} whose left and right edges in the input morphological representation do not correspond to the left and right edges of a constituent of type ω in the output phonological representation, such that N^{max} is a N that is not immediately dominated by another N (cmp. to Selkirk & Shen 1990). weight = 50

The computations of the prosodic parse of both compound structures are presented in tableaux (12) and (13). We begin with the computation of subcompounds in (12), the input morphological structure is that in (9b). Three output candidates are considered, the first in (12a) is the most faithful parse, where all N constituents are parsed into prosodic words, such that their edges match the edges of the prosodic words. This leads to a recursive prosodic word which violates the constraint NONREC and to a harmony score of -9. The more harmonic output candidate is (12b), which parses the entire structure into a single prosodic word. This output incurs two violations of $\text{MATCH}(\text{N}, \omega)$ due to the edges of two N constituents not matching the edges of any prosodic words ω in the output. Even with these two violations, candidate (12b) is most harmonic with its harmony score of -8 because of the low weight of $\text{MATCH}(\text{N}, \omega)$. Compare this to output (12c), which avoids one $\text{MATCH}(\text{N}, \omega)$ violation by parsing two N constituents into two prosodic words. Doing this incurs a fatal violation of high-weighted $\text{MATCH}(\text{N}^{\text{max}}, \omega)$ and a harmony score of -54.

(12) *Prosodic parsing of a subcompound:*

$[\text{N N N}]$		$\text{MATCH}(\text{N}^{\text{max}}, \omega)$ 50	NONREC 9	$\text{MATCH}(\text{N}, \omega)$ 4	$\mathcal{H}$
a.	$(\omega (\omega \text{N}) (\omega \text{N}))$		1		-9
 b.	$(\omega \text{N N})$			2	-8
c.	$(\omega \text{N}) (\omega \text{N})$	50		1	-54

In co-compounds, the computation favours a different pars given the difference in morphological structure. In subcompounds, a single prosodic word is most harmonic. In co-compounds, parsing two prosodic words is most harmonic, shown by candidate (13c) in the tableaux below. Crucially, (13c) does not violate any of the constraints presented in (11). It satisfies all MATCH constraints by parsing the two available Ns into individual prosodic words. It also satisfies NONREC by not exhibiting any recursive prosodic structure. In comparison to this, the outputs in (13a) and (13b) both violate MATCH(ω ,N) by virtue of parsing a prosodic word which does not match the edges of any N node, this becomes fatal in both cases⁵.

(13) *Prosodic parsing of a co-compound:*

[& N & N]	MATCH(ω ,N) 10	NONREC 9	MATCH(N, ω) 4	$\mathcal{H}$
a. (ω (ω N) (ω N))	1	1		-19
b. (ω N N)	1		2	-18
 c. (ω N) (ω N)				0

The Harmonic Grammar (HG) computations in (12) and (13) show that the different stress domains follow from a difference in morphological structure and MATCH constraints. Thus, the prosodic domains of simplex compounds can be computed in a parallel HG. This demonstrates that sub- and co-compounds need not be distinguished stratically. Without a stratal distinction between sub- and co-compounds the loop loses its function.

In addition to explaining the different stress domains of Malayalam compounds, the prosodic parse can also explain the application and non-application of lengthening at the compound boundary that is briefly mentioned in section 1. While we cannot do full justice to the details of the lengthening effects at compound boundaries, it will suffice to mention that the non-application of lengthening can be treated as a reflex of the prosodic structure in co-compounds. We assume that stem-initial gemination is

blocked due to the compound's internal prosodic word boundary. This boundary is not present in the prosodic structure of subcompounds and therefore, a consonant geminate freely surfaces in subcompounds. For a detailed analysis of the lengthening effects see Author & Author (2024). In the next section, we show that the prosodic domains in complex compounds are computed in parallel by implementing the same constraints that derive the prosodic structure of simplex compounds. This indicates further that the loop is an unnecessary device, as it is not required to derive the stress domains of complex compounds.

2.2 Stress domains in complex compounds

Mohanan (1986) gives a handful of examples of complex compounds, where either a subcompound is embedded in a co-compound, as in example (14) or a co-compound is embedded in a subcompound in example (15). In the example (14), the nouns *wife* and *sister* form a subcompound that is translated as *wife's sister*. This subcompound is then co-compounded with the noun *lady love*, which results in the complex compound translated as *lady love and wife's sister*. The stress domains in this example are transparent. The subcompound *wife's sister* supplies a single stress domain, as indicated by the single primary stress on the first long syllable of *wife*. Additionally, each head in the co-compound, *lady love* and *wife's sister*, constitutes its own independent stress domain, again indicated by the single primary stress on *lady love* and on the subcompound *wife's sister*.

- (14) [(kà:mukí:)(b^hà:ɽja:sahó:ɽaɽi)ma:ɽə]
 ka:muki- b^ha:ɽja- saho:ɽaɽi -ma:ɽ
 lady.love- wife- sister -PL
 ‘lady love and wife's sister’ [co-compound[subcompound]](p.123)

In the example (15) the nouns *caste* and *religion* form a co-compound. This co-compound is then subcompounded with the noun *hatred* to form the complex compound

hatred of religion and caste. Interestingly, each noun in the complex compound surfaces with its own primary stress, which indicates that each noun is parsed into a separate prosodic word. Thus, somewhat unexpectedly, the subcompound, which embeds a co-compound, does supply a single stress domain.

- (15) [(jà:ti̯)(màtá)(widwè:şám)]
 ja:ti̯-maťam-widwe:şam
 caste-religion-hatred
 ‘hatred of religion and caste’ [[co-compound]subcompound] (p.123)

Mohanan’s (1986) stratal architecture derives the stress domains of (15) with the assumption of the loop. His analysis is sketched in (16) below.

- (16) *Sketch of Mohanan’s (1986) analysis of (15)*
- | | |
|------------------------------------|--|
| str 3 subcompounding: N/A | [ja:ti̯] [maťam] [widwè:şam] |
| str 3 phonology: stress assignment | [jà:ti̯] [màtám] [widwè:şám] |
| str 4 co-compounding | [[jà:ti̯] [màtám]] [widwè:şám] |
| str 4 phonology | [[jà:ti̯] [màtám]] [widwè:şám] |
| loop to str 3 subcompounding | [[jà:ti̯màtá] [widwè:şám]] |

According to Mohanan, subcompounding and co-compounding happen at distinct strata where subcompounding precedes co-compounding. Thus in the derivation of (15), the nouns’ first visit to stratum 3 cannot yet yield a subcompound, as the embedded co-compound has not yet been built. Nevertheless, all nouns undergo the phonology of stratum 3, which assigns stress. Because none of the nouns have been compounded to form a larger domain, all nouns individually receive primary and secondary stresses. Stratum 4 morphology follows, where a co-compound can be built which undergoes stratum 4 phonology⁶. After stratum 4, with the help of the loop, the grammar can revisit stratum 3. At this stage, the subcompound can be built. However, it is too late for

the subcompound to be treated as a single domain for stress. What is essential to Mohanan's (1986) analysis of co-compounds embedded in subcompounds is that stratum 3 phonology (stress assignment) applies before stratum 3 morphology (subcompounding). This is only possible due to the mechanism of the loop. It ensures that under specific morphological circumstances, stratum 3 morphology is revisited after stratum 3 and 4 phonology has already been applied, which leads to the intransparent stress domains in complex compounds in (15).

Mohanan (1986) relies on a cyclic analysis where morphology and phonology can interleave and strict ordering of strata is relaxed. We assume a parallel prosodification of all sub- and co-compounds and by extension complex compounds. The intransparent stress effects in (15) seem to be a knock-down argument against a parallel theory to Malayalam compound morphophonology. However, we exemplify that the stress domains follow from our model of prosodification owing to the computation of Harmonic Grammar. First, we present the prosodic parsing of a co-compound that is embedded in a subcompound, à la (15) with the structure $[_N [\& N \& N] N]$. The related tableau is presented in (17). The most harmonic candidate is (17d), which is the most faithful parse of the complex compound, where each N is parsed such that its edges coincide with the edges of a prosodic word ω , yielding a recursive prosodic word. While this candidate violates NONREC, this violation is not fatal. This is a crucial departure from the derivation of simple subcompounds, where precisely this violation of NONREC ends up being fatal, see tableau (12) in the previous section. The crucial difference to simple compounds is the number of violations of $\text{MATCH}(N, \omega)$ that the alternative parse in candidate (17b) receives. (17b) parses the entire complex compound into a single prosodic word. By doing so, it incurs three violations of $\text{MATCH}(N, \omega)$ because there are three nominal elements involved in the morphological structure whose edges do not coincide with the edges of prosodic words. Compare this to the two nominal elements in the simple subcompound, leading to only two violations of $\text{MATCH}(N, \omega)$. Thus, in complex compounds, multiple violations of the low-weighted constraint $\text{MATCH}(N, \omega)$ lead to a

gang effect (Pater 2009, 2016), which worsens the harmony score of candidate (17b) to -12. Candidate (17d) only violates higher-weighted NONREC once, which leads to a more harmonic score of -9. Other prosodic parses that are presented by candidates (17a) and (17c) that avoid MATCH(N, ω) violations, cause violations of higher-weighted constraints and therefore do not improve the harmony score enough to win against the most harmonic (17d).

(17) *Prosodic parsing of a co-compound embedded in a subcompound:*

$[_N [\& N \& N] N]$	$M(N^{\max}, \omega)$ 50	$M(\omega, N)$ 10	NONREC 9	$M(N, \omega)$ 4	$\mathcal{H}$
a. $(_{\omega} (_{\omega} N) (_{\omega} N) N)$			1	1	-13
b. $(_{\omega} N N N)$				3	-12
c. $(_{\omega} N) (_{\omega} N) (_{\omega} N)$	1			1	-54
 d. $(_{\omega} (_{\omega} N) (_{\omega} N) (_{\omega} N))$			1		-9

For completeness sake, the tableau of the computation of subcompounds embedded in co-compounds is presented in (18). Here, the parsing works parallel to the individual parsing of sub- and co-compounds. The most harmonic output (18d) parses the subcompound into a single prosodic domain, thereby the parse avoids a recursive prosodic structure while simultaneously parsing the maximal N into a prosodic word. Additionally, it parses the individual N s of the co-compound into separate prosodic words. Crucially, the more faithful parse in (18a) that matches all constituents of type N into prosodic words is less harmonic. Just as in simple subcompounds this parse incurs a fatal violation of NONREC. Importantly, this violation is avoided in (18d) without leading to the gang effect that was observed in tableau (17) because there are only two nouns that are part of the subcompound structure compared to the three nouns of the subcompound structure in (17).

(18) *Prosodic parsing of a subcompound embedded in a co-compound:*

$[\& N \& [N N N]]$	$M(N^{\max}, \omega)$ 50	$M(\omega, N)$ 10	NONREC 9	$M(N, \omega)$ 4	$\mathcal{H}$
a. $(\omega N) (\omega (\omega N) (\omega N))$			1		-9
b. $(\omega N N N)$		1		2	-18
c. $(\omega N) (\omega N) (\omega N)$	1		1	2	-67
 d. $(\omega N) (\omega N N)$				2	-8

The computations of this and the previous section present a unified analysis of stress in Malayalam in parallel Harmonic Grammar that makes use of prosodic structure to arrive at the relevant prosodic domains for stress. Neither the parsing of simplex compounds nor that of complex compounds necessitates a stratal distinction between the compounds of Malayalam. Without this stratal distinction, there is no need for a loop.

3 Phonological opacity in co-compounds

Mohanan (1986) presents two cases of phonologically opaque surface forms which appear in co-compounds but not in subcompounds. Our assumption is that all compound types are derived at the same stratum, the stem stratum. This assumption conflicts with the observed opaque phonology in co-compounds for two reasons. The first reason is that the phonology of subcompounds and co-compounds should not be different. Given that both compounding types undergo the same stem-level phonology, phonological effects in co-compounds should be the same as in subcompounds. The second reason is that the opaque forms in co-compounds would constitute a case of intrastratal opacity, given that co-compounding happens in the first stratum. Kiparsky (2000) and Jaker & Kiparsky (2020) claim that intrastatal opacity is impossible under the assumptions of a stratal OT framework.

The first issue is immediately refuted under the assumption that prosodic structure causes different phonological effects in co-compounds compared to subcompounds despite their affiliation to a single stratum. The second issue is more problematic for our

minimized stratal proposal. Yet, we show that our stratal architecture can account for both opaque interactions in co-compounds under the assumptions of the established prosodic difference in compounds and a phonological cycle instantiated by word level phonology which follows stem level phonology.

3.1 Case 1: Final vowel lengthening and stress assignment

The first case of opacity presented by Mohanan (1986) concerns the process of final vowel lengthening and its interaction with stress assignment. Final vowel lengthening occurs in all types of nominal compounds and is described as follows: At the boundary between two compounds, a final vowel of the first stem is lengthened if the second stem has Sanskrit origin⁷. Example (19c) is the subcompound of the noun *bride* and *house*. In isolation, the noun *bride* surfaces with a short final vowel, see (19a). In the subcompound structure in (19c), the final vowel is lengthened. This lengthening interacts with stress placement. It attracts primary stress, boxed for convenience, because the lengthened vowel is in the second syllable of the word and the first syllable is short.

- | | | | |
|---------|----------|----|---|
| (19) a. | [wàḍʰú] | c. | [(_ω wàḍʰ ^h ùː grəhám)] |
| | ‘bride’ | | wàḍʰu-grəhám |
| | b. | | bride-house _[SANSKRIT] |
| | [grəhám] | | ‘bride’s home’ [subcompound] (p.120) |
| | ‘house’ | | |

To arrive at (19c) in a rule-based framework, the rule for final vowel lengthening must precede that of stress placement. This ensures that the lengthened vowel attracts stress. This interaction is one of shifting (Rasin 2022), where the application of the first rule influences the way the second rule is applied.

In co-compounds, the shifting interaction is not available. This becomes evident in example (20c) where the nouns *poet* and *poetess* are co-compounded. The final vowel of the first stem *poet* lengthens but does not surface with the expected primary stress.

The lengthened vowel in (20c) surfaces with a H tone instead.

- | | | | | |
|------|----|-------------|----|--|
| (20) | a. | [kàwí] | c. | [(_ω kàw[í:])(_ω kàwajitri)ka]ə] |
| | | ‘poet’ | | kawi-kawajitri-ka] |
| | b. | [kàwajitri] | | poet-poetess _[SANSKRIT] -PL |
| | | ‘poetess’ | | ‘poets and poetesses’ [co-compound] (p.121) |

The surface form in (20c) follows from rule ordering such that stress assignment precedes final vowel lengthening. This ordering leads to a countershifting interaction (Rasin 2022) where the second rule is expected to affect the way the first rule applies, but it ends up not doing so.

Thus, the rule ordering of final vowel lengthening and stress assignment is different depending on what type of compound is being dealt with. The conflict in rule ordering is resolved by Mohanan’s (1986) stratal distinction between the compounds, sketched in (21). At stratum 3, where nouns are subcompounded, final vowel lengthening applies before stress assignment. At stratum 4, where nouns are co-compounded, final vowel lengthening applies again and is thus ordered after stress assignment (of stratum 3 phonology).

(21) *Rule ordering to account for opacity:*

	Morphology	Phonology
stratum 3	subcompounding	1. final vowel lengthening 2. stress assignment
stratum 4	co-compounding	final vowel lengthening

Mohanan’s (1986) analysis relies on the stratal distinction between subcompounds and co-compounds. However, in order to arrive at the surface structures in (19c) and (20c) a stratal distinction is not necessary. The difference in phonological opacity follows directly from the proposed prosodic difference of the compounds and an additional

constraint, NON-FINALITY(FT'), on the placement of primary stress in prosodic words, defined in (22).

- (22) NON-FINALITY(FT'): Assign a violation to every rightmost foot head that is final in a prosodic word (Ito & Mester 2016). weight = 50

This constraint is weighted high throughout the grammar of Malayalam and penalises any foot head that is rightmost in the prosodic word. Recall that the final lengthened vowel in a co-compound structure is in exactly this position, see (23a). In subcompounds, the final lengthened vowel is not in a rightmost prosodic word final position, see (23b), therefore primary stress can be realised on the lengthened vowel.

- (23) a. $[(\omega \text{ kàw} \boxed{\text{í}})](\omega \text{ kàwajitri})\text{ka}[\emptyset]$ b. $[(\omega \text{ wad}^h \boxed{\text{ù}})]\text{gɾəhám}]$

The following tableau in (24) shows the computation of the prosodic structure and its interaction with stress placement and final vowel lengthening for co-compounds. We assume that the lengthening arises by the introduction of a floating mora between the two nouns⁸. Candidate (24a) integrates the floating mora, thereby creating the final lengthened vowel. However, primary stress does not fall on the lengthened vowel due to NON-FINALITY(FT'). Candidate (24c) realises primary stress on the long vowel, thereby incurring a fatal violation of high-weighted NON-FINALITY(FT') constraint. Alternatively, the prosodic parse can be changed to avoid a prosodic word boundary between the two compounded nouns. This is indicated by the less harmonic candidate (24b), which violates MATCH(ω , N) by parsing a prosodic word which does not match the edges of an N constituent. A violation of NON-FINALITY(FT') is also averted by not integrating the floating mora in candidate (24d) but doing so fatally violates *FLOAT. Thus, candidate (24a) is most harmonic despite not abiding by the weight-to-stress principle, which we argue is active in Malayalam and represented by the constraint WTS. The most harmonic candidate violates this constraint but due to the low weight of WTS, the violation is not fatal.

(24) *Non-Finality in co-compounds:*

	[& [N kawi] (μ) [N kawajitri]]	NON-FIN(FT')	MATCH(ω , N)	*FLOAT	WTS	$\mathcal{H}$
		50	10	7	3	
☞ a.	(ω kàwí:) (ω kàwajitri)				1	-3
b.	(ω kawì:kawajitri)		1			-10
c.	(ω kawì:) (ω kàwajitri)	1				-50
d.	(ω kàwí) (μ) (ω kàwajitri)			1		-7

Independent evidence for the NON-FINALITY(FT') constraint to hold in Malayalam comes from bisyllabic words that underlyingly surface with a long vowel in word final position. Take the word *what* in the sentence (25) below. Its final vowel does not surface with primary stress despite canonically being in the correct position. Under the assumption that *what* is parsed into a prosodic word, the final long vowel is rightmost in the prosodic word and thus, according to NON-FINALITY(FT'), cannot be the head in a foot and therefore does not receive primary stress.

(25) mà:liní mà:lá èndé: cèjtató

Malini necklace what did

What did Malini do with the necklace?(pp.113-114)

Summing up, this first case of phonological opacity in Malayalam co-compounds does not provide evidence for a stratal distinction between sub- and co-compounds. The constraint NON-FINALITY(FT') and the prosodic parse of co-compounds conspire to compute a surface form that seems to be a case of intrastratal opacity. However, the computation remains fully parallel.

3.2 Case 2: Vowel sandhi and stress assignment

The second case of phonological opacity concerns the interaction of vowel sandhi and stress assignment. Mohanan observes that if two vowels become adjacent at a compound boundary and the nouns in the compound have Sanskrit origin, multiple vowel

sandhi processes occur to resolve the vowel hiatus. Table I shows all possible vowel combinations and their surface realisation.

V1	V2	surface
a	i	ee
a	u	oo
a	a	aa
i	i	ii
aa	au	au
i	a(i/u)	ja(i/u)
u	a(i)	wa(i)

Table I: Summary of vowel sandhi in Malayalam

In almost all cases, a long vowel results, usually from fusion of the two adjacent vowels. It is this resulting long vowel, which may interact with stress assignment in Malayalam compounds. Similar to the interaction between vowel final lengthening and stress assignment, the surface long vowel that results from vowel sandhi in subcompounds attracts primary stress if it is in the second syllable of the compound. This is exemplified by the subcompounding of *water*⁹ and the vowel-initial noun *desire* in (26).

- (26) [jal^hè:cc^ha]
jala(m)-icc^ha
water-desire
‘thirst’ (subcomp) (p.120)

Analogous to the interaction of vowel final lengthening and stress assignment, the surface realisation in (26) is derived in a rule-based framework if the rules of vowel sandhi apply before the rules of stress assignment. This ensures a shifting interaction where vowel sandhi can influence the result of stress assignment.

In co-compounds, no shifting interaction is found. Instead, the fused vowel surfaces with a falling contour, see (26). This follows if stress assignment applies before vowel sandhi in co-compounds. By applying stress assignment first, the final syllable of the proper noun *Tara* receives the boundary H tone and the first syllable of the proper noun

Indran receives the primary stress realised as an L pitch. Fusion of the vowels leads to fusion of the tones and, therefore, a falling pitch contour. If the rules were in the opposite order, as they are for subcompounds, then the resulting surface form would be [t̪àːt̪ éːndr̪ánmaːɾə] where secondary stress surfaces on the fused long vowel. Evidently, this is not the surface form in co-compounds.

(27) [t̪àːt̪ éːndr̪ánmaːɾə]

t̪aːɾa-indr̪an-maːɾə

Tara-Indran-PL

‘Tara and Indran’ (co-comp) (p.121)

As was the case in the previous section, analysing phonological opacity in co-compounds in a rule-based framework necessitates different rule orderings for sub- and co-compounds, which suggests that co-compounds undergo a different stratum of phonology compared to subcompounds. However, with the power of prosodic structure and a minimised stratal architecture, the patterns in (26) and (27) can be derived while still allowing sub- and co-compounds to be built at the same morphological stratum, the stem stratum. In a nutshell, what is crucial to our analysis is that nouns in co-compounds are parsed as separate prosodic words. This parsing, we argue, blocks vowel sandhi at the stem stratum. Then, at the word stratum, the entire co-compound plus any added inflectional affixes are parsed as a recursive prosodic word, which provides the correct environment for vowel sandhi. It follows that vowel sandhi in co-compounds is only facilitated at the word stratum after stress has been assigned at the stem stratum.

To derive the different effects of vowel sandhi in sub- and co-compounds, we introduce the high-weighted constraint in (28) below. This constraint is penalised by adjacent vowels in the same prosodic word.

(28) $*_{(\omega} V.V)$: Assign a violation to every sequence of (nonidentical) vowels in the same prosodic word. weight = 50

At the phonology of the stem stratum, a vowel hiatus in a subcompound structure is resolved due to the parsing of the subcompound as a single prosodic word and the high weight of $*(_{\omega} V.V)$, see the sketched analysis in (29) below.

(29) Derivation of vowel sandhi and stress shifting in subcompounds:

Stem-level: Subcompounding $[_N [_N \text{jala(m)}] [_N \text{icc}^h\text{a}]]$

Stem-level Phonology: $(_{\omega} \text{jal} \boxed{\text{è:}} \text{cc}^h\text{a})$

A vowel hiatus at the co-compound boundary is not in the same prosodic word, due to the parsing of each noun into independent prosodic words. Thus, the constraint $*(_{\omega} V.V)$ is vacuously satisfied by co-compounds at the stem-level. Therefore, at the stem level the hiatus remains unresolved, yet stress domains are already established. At the word stratum, inflectional affixes enter the morphological computation. These affixes are prosodified at word stratum phonology, resulting in the entire compound word being parsed into a recursive prosodic word. Due to this parse, the constraint $*(_{\omega} V.V)$ is active and vowel sandhi applies, which fuses the H boundary tone on the final syllable of the first noun and the L pitch of the primary stress on the first syllable of the second noun into a falling contour. This cyclic derivation is sketched in (30).

(30) Cyclic derivation of vowel sandhi and stress countershifting in co-compounds:

Stem-level: Co-compounding $[_{\&} [_N \text{tà:}^h\text{a}] [_N \text{ìndr}^h\text{án}]]$

Stem-level Phonology: $(_{\omega} \text{tà:}^h\text{a}) (_{\omega} \text{ìndr}^h\text{án})$

Word-level: Inflectional affixation $[[_{\&} (_{\omega} \text{tà:}^h\text{a}) (_{\omega} \text{ìndr}^h\text{án})] -\text{ma:}^h\text{ɿ}]$

Word-level Phonology: $(_{\omega} (_{\omega} \text{tà:}^h\text{a}) (_{\omega} \text{èndr}^h\text{án}) \text{ma:}^h\text{ɿ})$

$[\text{tà:}^h\text{é:ndr}^h\text{ánma:}^h\text{ɿ}]$

This second case of phonological opacity thus constitutes interstratal opaque phonology, which cannot be handled intrastratally, as opposed to the first case of opacity. Mohanan (1986) takes this as evidence that the two compound structures must be built at distinct strata, thereby undergoing distinct phonologies. We show that the stratal dis-

tion is not necessary. Instead, the phonological difference between the compounds is prosodic. In co-compounds, the prosodic parse postpones vowel sandhi until the next stratum, which results in the opaque surface form. In subcompounds, the prosodic parse immediately facilitates vowel sandhi at stem level phonology, which leads to the transparent shifting interaction.

4 Conclusion

In this paper, we provide a new perspective on Malayalam compound morphophonology which argues against Mohanan’s (1986) stratal architecture and in particular argues that a stratal looping mechanism is avoidable. The central insight we contribute is that the morphological structure of subcompounds and co-compounds force different prosodic parses. The difference in prosodic parsing reflects the difference in phonological surface forms. We demonstrate that stress domains in simplex and complex compounds follow from the different prosodic parses of sub- and co-compounds. Additionally, we show that opaque phonology in co-compounds follows from two assumptions. First, a minimal stratal architecture consists of a stem-, a word- and a postlexical stratum, where all compounding takes place at the stem stratum. By virtue of this assumption, interstratal opacity is evoked by additional prosodification at the word stratum. Second, the inclusion of prosodic structure which matches morphological structure demonstrates that intrastratal opacity at the stem stratum is merely a quirk of the prosodic parsing of co-compounds and a NON-FINALITY constraint. Thus, we have shown that overwhelming empirical evidence pointing towards the need to abandon level ordering by introducing a loop is explained by prosodic structure computed by MATCH constraints in Harmonic Grammar and a minimised, restrictive stratal architecture that abides by strict level ordering.

References

- Author, A. & B. Author. 2024. [Masked for anonymous review]. *In preperation*.
- Bermúdez-Otero, Ricardo. 2018. Stratal Phonology. In S. J. Hannahs & Anna R. K. Bosch (eds.), *The Routledge handbook of phonological theory*, 100–134. Abingdon: Routledge.
- Di Sciullo, Anna Maria. 2005. Decomposing compounds. *Skase Journal of theoretical linguistics* 2(3). 14–33.
- Di Sciullo, Anna Maria. 2011. Why are compounds a part of human language? A view from asymmetry theory. In Rochelle Lieber & Pavol tekauer (eds.), *The Oxford Handbook of Compounding*, 145–177. Oxford: Oxford University Press.
- Inkelas, Sharon. 1989. *Prosodic constituency in the lexicon*. Stanford, CA: Stanford University dissertation.
- Inkelas, Sharon. 2014. *The interplay of morphology and phonology*. Oxford: Oxford University Press.
- Ito, Junko & Armin Mester. 2016. Unaccentedness in Japanese. *Linguistic Inquiry* 47(3). 471–526.
- Jaker, Alessandro & Paul Kiparsky. 2020. Level ordering and opacity in Tetsót'né: a Stratal OT account. *Phonology* 37(4). 617–655.
- Kiparsky, Paul. 1982. Lexical phonology and morphology. In *Linguistics in the morning calm*, 3–91. Seoul: Hansin.
- Kiparsky, Paul. 2000. Opacity and cyclicity. *The Linguistic Review* 17. 351–367.
- Legendre, Geraldine, Yoshiro Miyata & Paul Smolensky. 1990. Can Connectionism Contribute to Syntax? Harmonic Grammar, with an Application. *CLS* 26(1). 237–252.
- Mohanan, K. P. 1982. *Lexical phonology*. MA: Cambridge, Massachusetts Institute of Technology dissertation.

- Mohanan, K. P. 1986. *The theory of lexical phonology*. Dordrecht: D. Reidel Publishing Company.
- Mohanan, K. P. 1996. 2. The Organization of the Grammar. In John A. Goldsmith (ed.), *The handbook of phonological theory*. Blackwell Publishing. Blackwell reference Online. 31 December 2007. http://www.blackwellreference.com/subscriber/book?id=g9780631201267_9780631201267.
- Nóbrega, Vitor A. 2020. No escape from categorization: an insiders view of compounds. *Ilha do Desterro* 73(3). 103–126.
- Orgun, Cemil Orhan. 1996. *Sign-based morphology and phonology with special attention to Optimality Theory*. Berkeley, CA: University of California dissertation.
- Pater, Joe. 2009. Weighted constraints in generative linguistics. *Cognitive science* 33(6). 999–1035.
- Pater, Joe. 2016. Universal grammar with weighted constraints. In John J. McCarthy & Joe Pater (eds.), *Harmonic grammar and harmonic serialism*, 1–46. London: Equinox.
- Pesetsky, David. 1979. Russian morphology and lexical theory. *Ms., MIT*.
- Ralli, Angela. 2019. Coordination in compounds. In *Oxford research encyclopedia of linguistics*. Oxford University Press Online. accessed 26. September 2024. <https://doi.org/https://doi.org/10.1093/acrefore/9780199384655.013.564>.
- Rasin, Ezer. 2022. Shifting interactions and countershifting opacity: a note on opacity in Harmonic Serialism. *Linguistic Inquiry* 53(4). 836–851.
- Selkirk, Elisabeth & Tong Shen. 1990. Prosodic Domains in Shangahi Chinese. In Sharon Inkelas & Draga Zec (eds.), *The phonology-syntax connection*, 313–337. Chicago, University of Chicago Press.
- Selkirk, Elisabeth O. 1995. The prosodic structure of function words. In *University of massachusetts occasional papers 18: papers in optimality theory*, 439–469. GLSA, University of Massachusetts, Amherst.

- Selkirk, Elisabeth O. 2011. The syntax-phonology interface. In John Goldsmith, Jason Riggle & Alan C. L. Yu (eds.), *The handbook of Phonological Theory*, 435–484. Wiley Online Library.
- Sproat, Richard. 1985. *On deriving the lexicon*. Cambridge, MA: MIT dissertation.
- Sproat, Richard. 1986. Malayalam compounding: a non-stratum ordered account. In *Proceedings of WCCFL*, vol. 5, 268–288.
- Wälchli, Bernhard. 2005. *Co-compounds and Natural Coordination*. Oxford: Oxford University Press.

Notes

¹In a stratal OT framework there are no phonological rules, rather the ranking of constraints may change from stratum to stratum

²In this and all future examples in the paper, stress is indicated with its tonal correlates. Primary stress is represented as an L tone and secondary stress as an H tone.

³All page numbers in the examples refer to Mohanan (1986) if not otherwise indicated.

⁴Mohanan (1986) notes that stress in Malayalam does not manifest itself phonetically as a change in amplitude, duration or fundamental frequency. However, vowels that are unstressed, meaning vowels that are not pronounced at L pitch, are reduced or deleted entirely. Vowels that are pronounced at L pitch can never be reduced or deleted.

⁵Output (13b) also violates high-weighted MATCH($N^{\max}, \omega$) twice, which immediately rules this candidate out.

⁶Stratum 4 phonology according to Mohanan (1986) includes compound vowel lengthening, nasal deletion and schwa epenthesis among other phonological processes. None of the processes are relevant in the example (15).

⁷If the second noun in a compound has Dravidian origin then a process of initial gemination applies. This was presented in example (2) in section 1, $\text{petti- patta:jam -ka} \rightarrow \text{petti} \boxed{\text{pp}} \text{attá:jáŋgalə}$. This process is restricted to subcompounds; it never occurs in co-compounds. This follows under our analysis using prosodic domains, if there is a positional restriction on gemination which disprefers a geminate to surface at the left edge of a prosodic word. For a concrete implementation of compound gemination and its restriction, we refer to Author & Author (2024)

⁸The floating mora can be analysed as a linking element, similar to the way gemination is also treated as a linking element in Author & Author (2024)

⁹The noun *water* is not vowel-final underlyingly. It has an underlying final nasal, which deletes if in a compound structure. Nasal deletion in compounds feeds vowel sandhi, as the example (26) indicates.